Acoustic Visual Guided Diffusion for Missing View Image Generation

IE 643 Course Project: Intensive Assessment (Stage 1)

Team: Natural Stupidity

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Code: github.com/siddheshsp0/ie-643-project · IIT Bombay, October 2026

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Problem statement

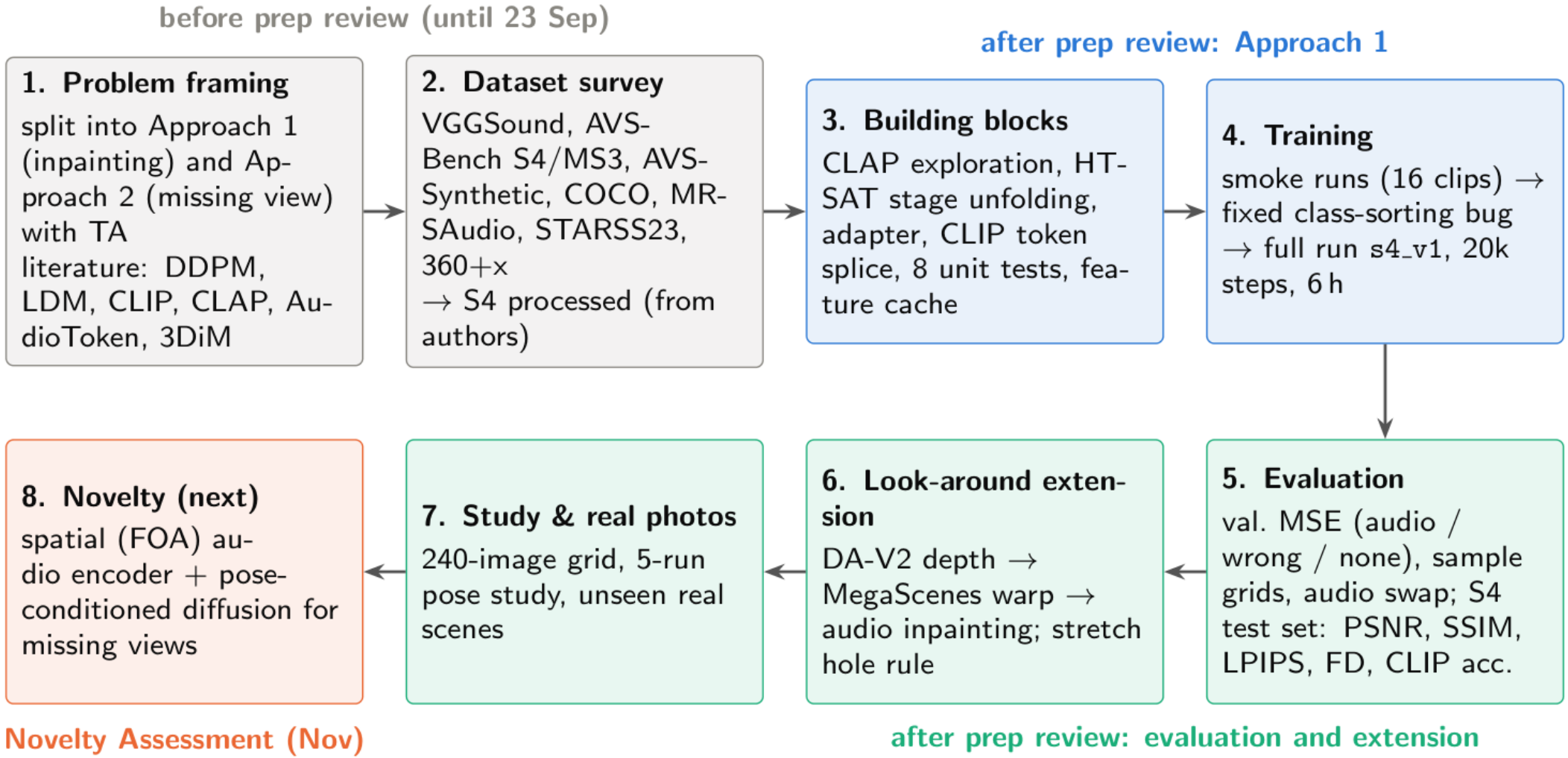
The task

- A camera misses an object (occlusion, limited field of view, darkness, sensor placement); a microphone still hears it.
- Given the **observed RGB image** and **acoustic evidence**, generate the missing region or unseen view.
- The **sound should decide what is drawn**, instead of the model guessing from RGB alone.
  - Example: a corridor camera cannot see around a corner, but footsteps say a person is there.

Our split (agreed with the TA)

- **Approach 1:** masked image + mask + sound → completed image (audio-conditioned inpainting).
- **Extension:** move the camera with a depth warp, then inpaint only the pixels the photo never saw.
- **Approach 2 → Novelty Assessment:** observed view + spatial (FOA) audio → unseen view, source at the right place.

Project workflow



Work done before the prep review (until 23 Sep)

Framing

- Split the task into Approach 1 (inpainting, mono audio) and Approach 2 (spatial audio, new view).

Literature

- DDPM, latent diffusion and SD inpainting, CLIP and CLAP, AudioToken.
- Novel-view diffusion: 3DiM, Zero-1-to-3, Free3D, MultiDiff, SV3D, CAT3D.

Proposed design

- Sound → CLAP → adapter → CLIP word space → frozen SD-1.5 inpainting U-Net.

Dataset survey

- Audio-visual: VGGSound, AVSBench S4 / MS3, AVS-Synthetic.
- Image + text: MS COCO captions.
- Spatial audio: MRSAudio, STARSS23, 360+x, SoundSpaces 2.0.
- **Chosen for Approach 1: AVSBench-S4** (frame + sounding-object mask + audio from the same video); processed data obtained from the AVSBench authors.

Work done after the prep review

Approach 1: audio-conditioned inpainting

- Explored CLAP embeddings and text-prompted SD inpainting.
- Multi-stage CLAP feature extractor (stage hooks, exact time un-folding), verified by unit tests.
- AudioAdapter (5.8M params) + CLIP token splice; 8 unit tests; CLAP features cached for all 4,932 clips.
- Smoke runs on 16 clips → found and fixed a class-ordering bug in the validation subset.
- Full training run **s4_v1**: 20k steps, ~6 h on one RTX A5000.

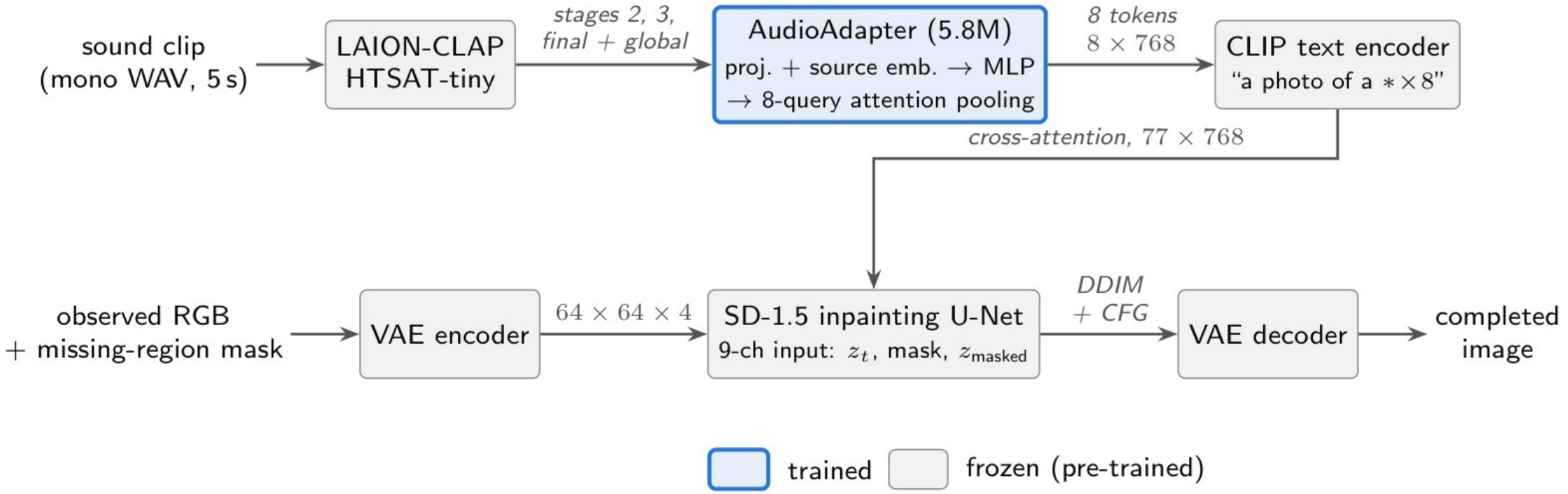
Evaluation and extension

- Validation with own / wrong / no audio; full S4 test set (739 clips) with PSNR, SSIM, LPIPS, FD-CLIP, CLIP accuracy and 3 baselines.
- **Look around occluders:** Depth Anything V2 + MegaScenes warp + our audio inpainting; new hole rule.
- 240-image grid, 5-run camera-pose study, unseen real photos.

Change of plan

- Skipped the caption pre-training stage; trained the adapter directly on the inpainting loss.

Approach 1: audio-conditioned inpainting



- **Trained:** AudioAdapter only (5.8M parameters). **Frozen:** CLAP, CLIP text encoder, VAE, U-Net (~860M).
- Audio becomes 8 tokens that fill the prompt **"a photo of a * * * * * * * *"**; CLIP then runs unchanged.
- U-Net input has 9 channels: noisy latent + mask + masked-image latent; the prompt enters by cross-attention.

AudioAdapter and training objective

AudioAdapter (AudioToken-style)

- Input: CLAP HTSAT stage 2 [64×384], stage 3 [32×768], final [32×768], global [512].
  - time kept, frequency averaged; 129 tokens in total
- Per-stage Linear → 768 + learned source embedding.
- MLP 768 → 1024 → 768 (GELU) on every token.
- 8 learned queries, 8-head attention pooling → **8 tokens × 768**.

Why

- Reuse SD's text space; few parameters for a small dataset.

Loss (per step)

$$L = \Sigma\, w\cdot(\varepsilon_\theta - \varepsilon)^2 \,/\, \Sigma\, w \;+\; 0.001 \cdot \text{mean}|\text{tokens}|$$

$$w = 1 + 2\cdot m \quad \text{(masked region weighted 3×)}$$

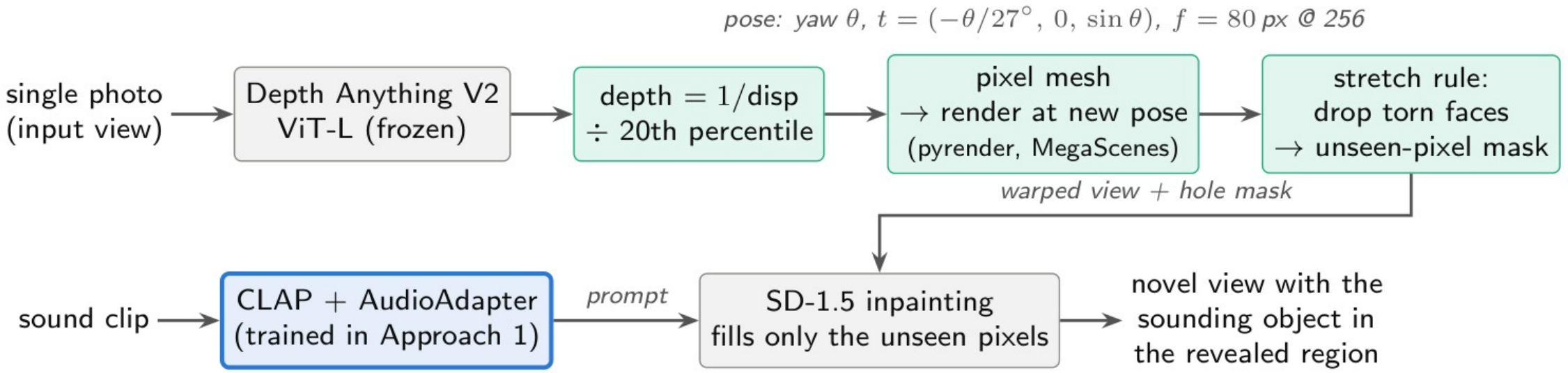
$$\varepsilon_\theta = \text{UNet}([z_t, m, z_{\text{masked}}], t, \text{CLIP}(\text{prompt} + \text{tokens}))$$

- 10% condition dropout → classifier-free guidance (scale 7.5).

Mask mix per sample

- 60% sounding-object mask: draw the object in context.
- 20% random box (10–50% of image): general fill.
- 20% full mask: whole scene from sound.

Extension: look around occluders



- **Depth:** Depth Anything V2 (ViT-L) disparity → depth = 1/disparity, divided by its 20th percentile.
- **Warp:** one mesh vertex per pixel, rendered with pyrender from the MegaScenes orbit (±15° yaw + translation).
- **Our stretch rule:** drop only mesh faces torn open by parallax, so holes = pixels never seen. MegaScenes' angle rule removes 19–25% of the image even with no camera motion.
- **Fill:** audio-conditioned SD inpainting on the dilated hole mask; seen pixels copied back unchanged.

Dataset and preprocessing

AVSBench-S4 (single source)	Train	Val	Test
5-s clips (23 classes)	3,452	740	740
Frames per clip (224×224, 1 fps)	5	5	5
Sounding-object masks	frame 1	all 5	all 5

Source

- AVSBench (Zhou et al., ECCV 2022); processed S4 data from the dataset authors.
- Look-around tests: S4 test frames, a phone photo and scene-classification images never used in training.

Image preprocessing

- Frames resized to 512×512 (bicubic), scaled to [−1, 1].
- Masks: nearest resize, dilated by 8 px; random horizontal flip.

Audio preprocessing

- 48 kHz mono; 5-s clip repeat-padded to CLAP's 10-s window.
- CLAP features cached once for all clips (fp16, 699 MB).

Splits

- Official train / val / test; class-balanced (round-robin) subsets for validation.

Experimental setup

Training (run s4_v1)	Value
Optimizer	AdamW, lr 8e-5, wd 0.01
Schedule	500 warm-up steps + cosine
Batch	4 × 2 accumulation = 8
Steps	20,000 (~46 epochs)
Grad. clip / cond. dropout	1.0 / 0.1
Precision	U-Net bf16 autocast + checkpointing
Hardware	1× RTX A5000 24 GB, peak 6.0 GB
Time	~0.94 steps/s, ~6 h

Validation

- 64 val clips (all classes), fixed noise and timesteps; masked MSE with own / wrong / no audio.

Test set (Approach 1)

- All 739 S4 test clips with a mask, DDIM 30 steps, CFG 7.5; same seed per clip for 4 conditionings: own audio, wrong audio, no audio, class-name text (oracle).
- Metrics: PSNR, SSIM, LPIPS, FD-CLIP, CLIP ViT-L/14 zero-shot top-1 (23 classes).

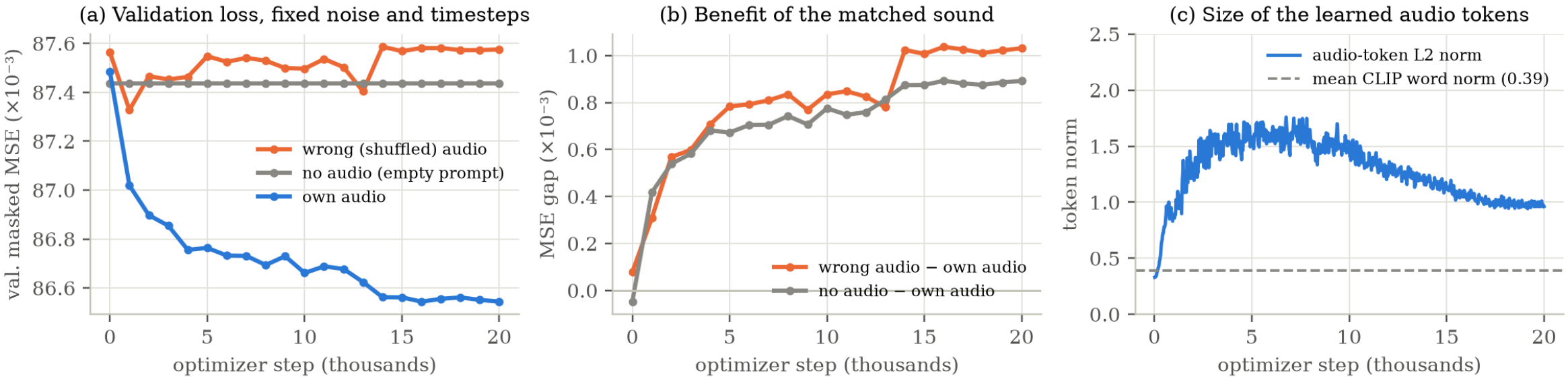
Look-around

- 8 scenes × ±15° × 5 conditionings × 3 seeds = 240 images; 4 more camera moves.

Software

- PyTorch 2.6, diffusers 0.40, transformers 5.17, laion-clap 1.1.7.

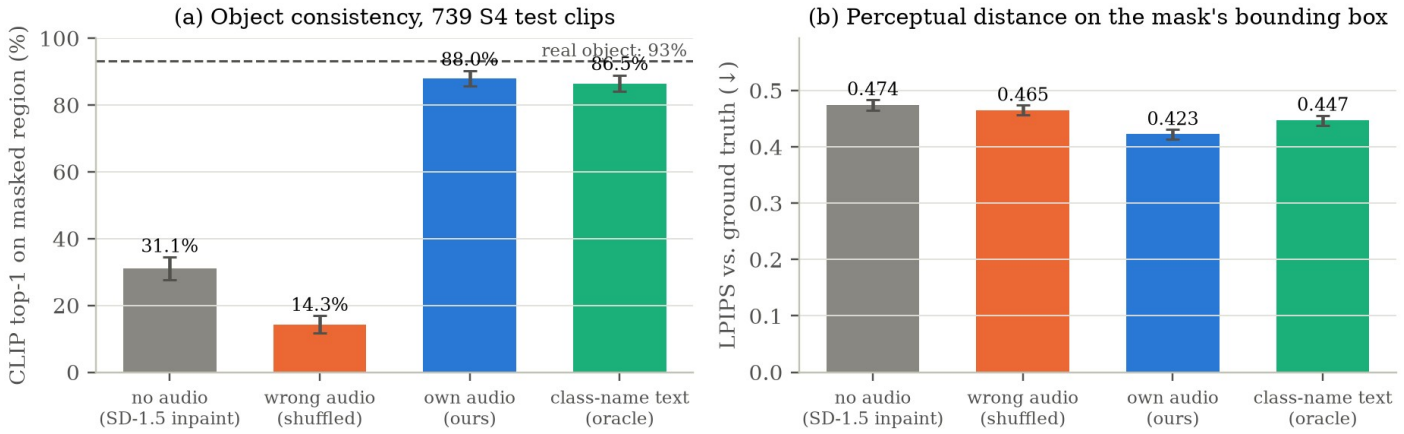
Results: training and validation



Val. masked MSE (step 20k)	Value
Own audio	**0.0865**
No audio (empty prompt)	0.0874
Wrong (shuffled) audio	0.0876

- Own audio < no audio < wrong audio, as required.
- Gap grows over the whole run → no overfitting; last checkpoint used.
- Token norm settles at 0.96 (CLIP words: 0.39).

Results: S4 test set (739 clips)

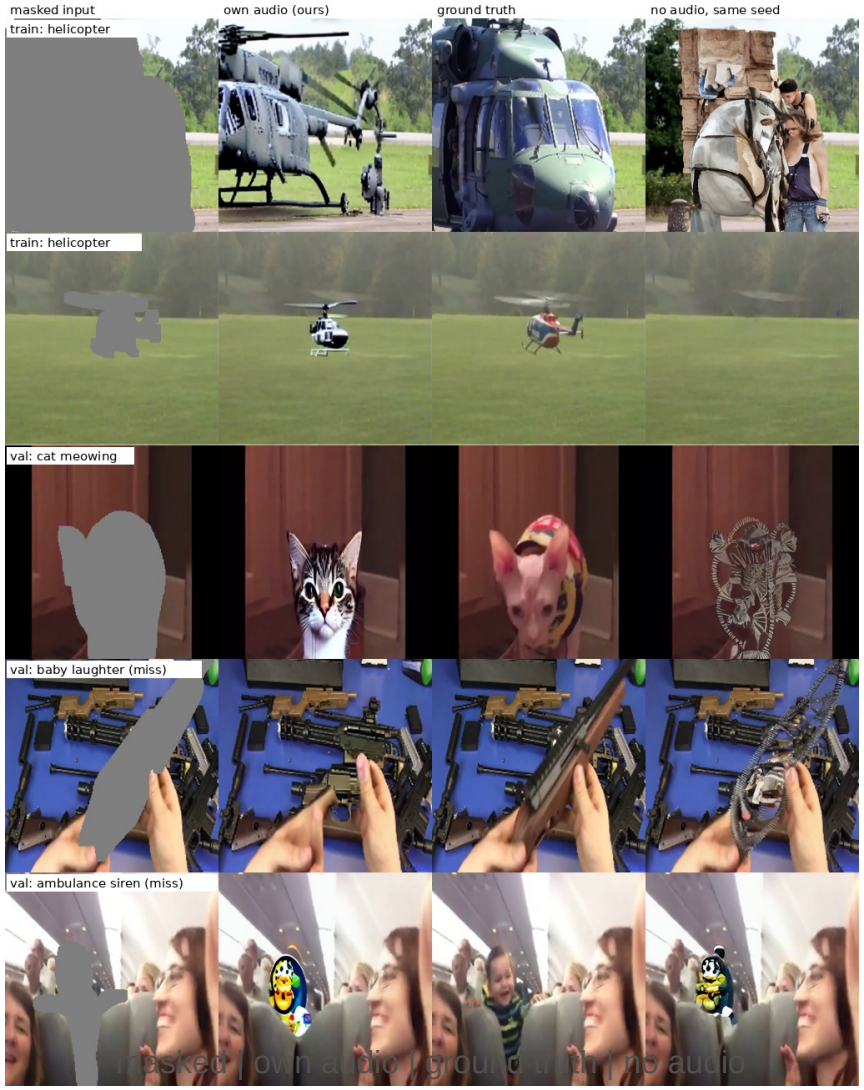


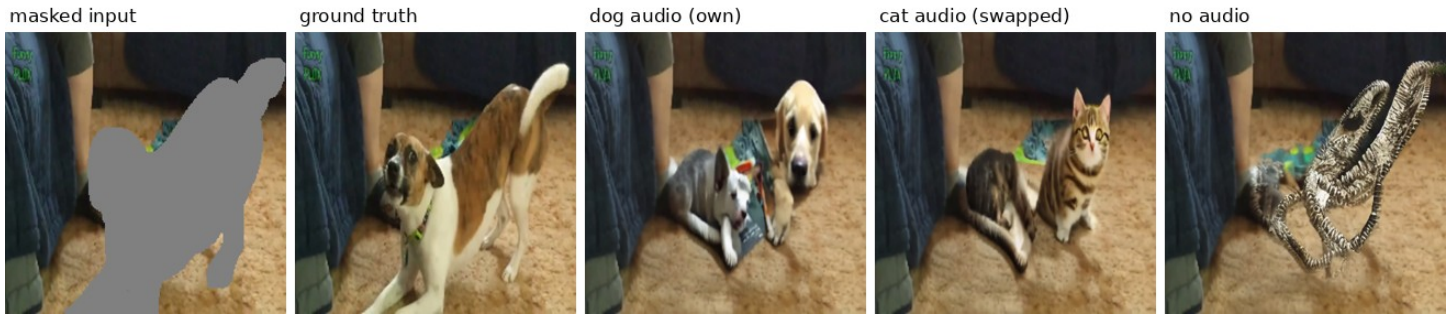
Conditioning	PSNR ↑	SSIM ↑	LPIPS ↓	FD-CLIP ↓	CLIP top-1 ↑
No audio (SD-1.5)	10.97	0.516	0.474	0.224	31.1%
Wrong audio	9.94	0.525	0.465	0.131	14.3%
Own audio (ours)	**9.97**	**0.519**	**0.423**	**0.080**	**88.0%**
Class-name text (oracle)	10.35	0.525	0.447	0.116	86.5%

What it shows

- Own audio: right object **88%** vs **31%** without audio (oracle text 86%, real object 93%).
- Wrong audio: the drawn object matches the **sound's class 48%** of the time, the image's class only 14%.
- Own audio has the lowest LPIPS (0.423) and FD-CLIP (0.080) of the four.
- Per clip, own audio raises P(true class) over no audio for 92% of clips.
- PSNR/SSIM favour bland fills; they cannot tell a plausible object from the true one.

Results: qualitative inpainting

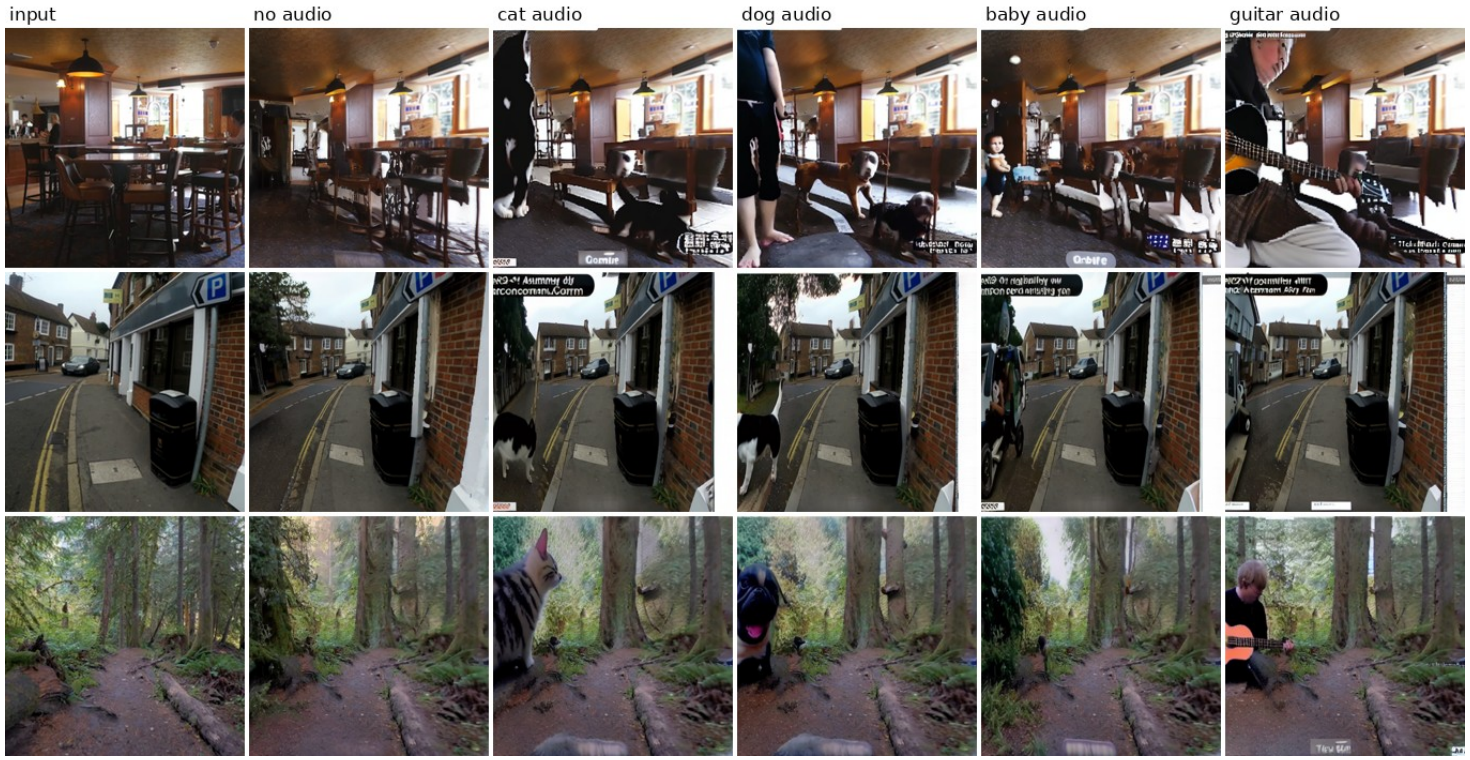




Unseen test clip: same frame, mask and seed; only the sound changes

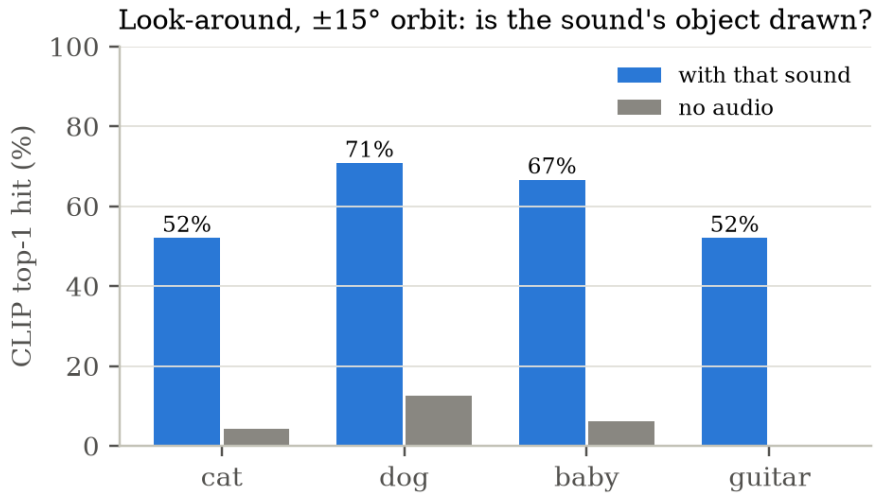
- Own sound → the right object (helicopter, cat, dog, toy gun).
- Same seed, no audio → clutter or unrelated objects.
- Swap dog audio for cat audio → cats instead of dogs: **the sound decides what is drawn.**
- Misses: baby laughter (cartoon toy), tiny masks such as a distant ambulance.

Results: look around occluders



Unseen real scenes, +20° orbit, seed 0 for all images

CLIP top-1 on generated pixels	with audio	no audio
S4 grid, ±15° orbit (240 images)	60%	6%
Best camera move, +20° orbit	83%	8%



- Paired win rate (audio > no audio): 83%.
- Larger revealed regions → stronger audio control.
- Thin gaps next to an object are filled with more of that object.
- Artifacts: garbled text and watermarks (YouTube training frames).

Conclusions

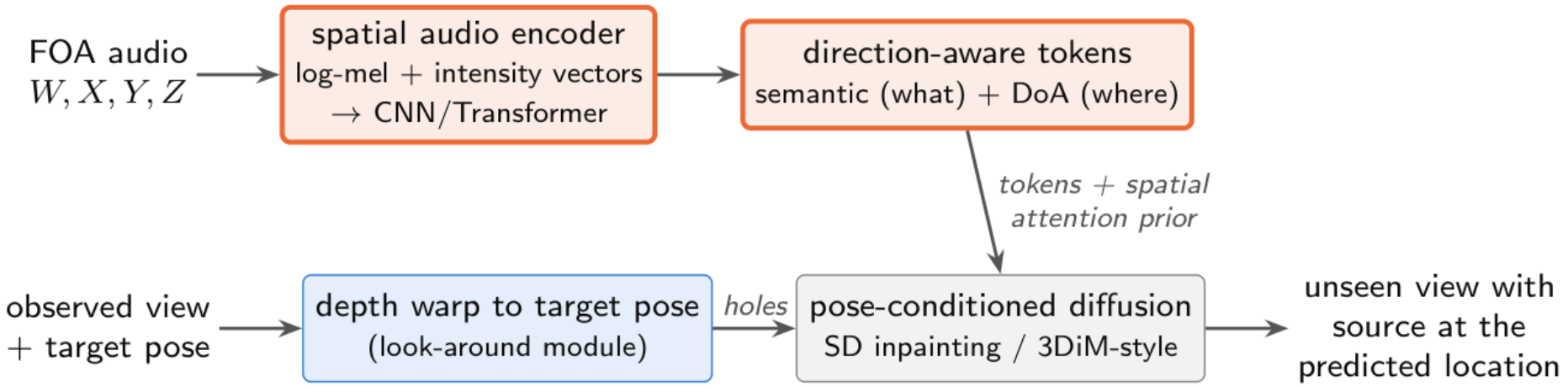
- A 5.8M-parameter adapter makes frozen SD-1.5 inpainting **follow sound**: own audio < no audio < wrong audio in validation loss, and the sound changes what is drawn.
- S4 test set (739 clips): right object 88% with own audio vs 31% without, on par with an oracle text prompt (86%); lowest LPIPS and FD-CLIP.
- The same model fills **regions a camera never saw** after a depth-based view change: 60% vs 6% (up to 83% at +20°).

Limitations

- 224-px frames → blurry fills; text/watermark artifacts; thin holes beside objects; mono audio gives **what**, not **where**.

Code: github.com/siddheshsp0/ie-643-project

Plan for Novelty Assessment: spatial audio → unseen view



- **Goal:** observed view + FOA audio (W, X, Y, Z) + target pose → unseen view with the source at the right place.
- **Spatial encoder:** log-mel + intensity vectors → tokens for what and where (direction of arrival).
- **Reuse:** look-around warp for the visible part, audio inpainting for the rest.
- **Data:** STARSS23 (FOA + 360° video + DoA labels), MRSAudio, SoundSpaces 2.0 (known geometry).
- **Experiments:** visual-only vs mono vs FOA conditioning.
- **Metrics:** image quality, object presence (CLIP), location error vs true direction.

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